ENTERPRISE ENGINEERING & PLATFORM ARCHITECTURE

EDWORLD CO

Comprehensive Platform Showcase, System Architecture, Execution Pipelines, and Multi-Tier Scalability Strategy.

Adaptivity Engine

Dynamic calibration of selfLevel vs effectiveLevel with automated difficulty nudging.

Minors Safety by Design

COPPA, GDPR-K, and DPDP Act compliance with strict privacy & zero appearance scoring.

AI Speech Diagnostics

Whisper speech-to-text, STAR framework evaluation, WPM cadence, and ATS resume audits.

VERSION & BUILD
Production Release v2.4

ARCHITECTURE FRAMEWORK
Next.js 14 + Supabase + Firebase + Claude

REPORT DATE
September 2026

1. Executive Summary & Core Mission

Edworld Co is an AI-powered personalized career-readiness, skills-growth, and institutional workforce development platform. Unlike traditional one-size-fits-all career portals, Edworld Co dynamically adjusts what it teaches, how it scores, and what opportunities it presents based on the candidate's exact **age band**, **verified skills**, and **demonstrated real-time performance**.

> **The Individual Adaptivity Principle**
> Every learner declares a baseline skill level (`selfLevel` : 0 = Foundational, 1 = Intermediate, 2 = Advanced) during onboarding. As they practice mock interviews, coding challenges, and verbal drills, the scoring engine calculates their calibrated `effectiveLevel` . Achieving ≥ 80% automatically nudges the difficulty level upward, serving progressively harder real-world challenges.

Target User Ecosystem

Junior Track (Ages 13–15)

Curiosity & foundational skills lab, microchip logic gates, money basics, ethical AI modules, and reflective journaling.

FOUNDATIONAL LAB

Explorer Track (Ages 16–18)

College & scholarship directory, SOP & essay coach, engineering portfolio starter kits, and verbal communication drills.

COLLEGE & PORTFOLIO

Career Track (Ages 18+)

3-template ATS resume builder, AI mock interview studio, verified jobs board, mentor directory, and 1-on-1 direct chat.

PLACEMENT & JOBS

Institutional & Enterprise Ecosystem

Employers & Talent HR

Job creation, skill-level matching engine, applicant tracking pipeline (ATS), and private recruiter evaluation notes.

Educators & Faculty

Virtual classroom creation, join codes, curriculum task assignments, and student AI reflection transparency logs.

Founders & Superadmins

Platform telemetry, real-time user metrics, content moderation queue, and immutable security audit logging.

2. Full System Architecture

The platform is built on a 5-tier cloud-native architecture designed for low-latency client interaction, secure multi-tenant isolation, and resilient serverless compute.

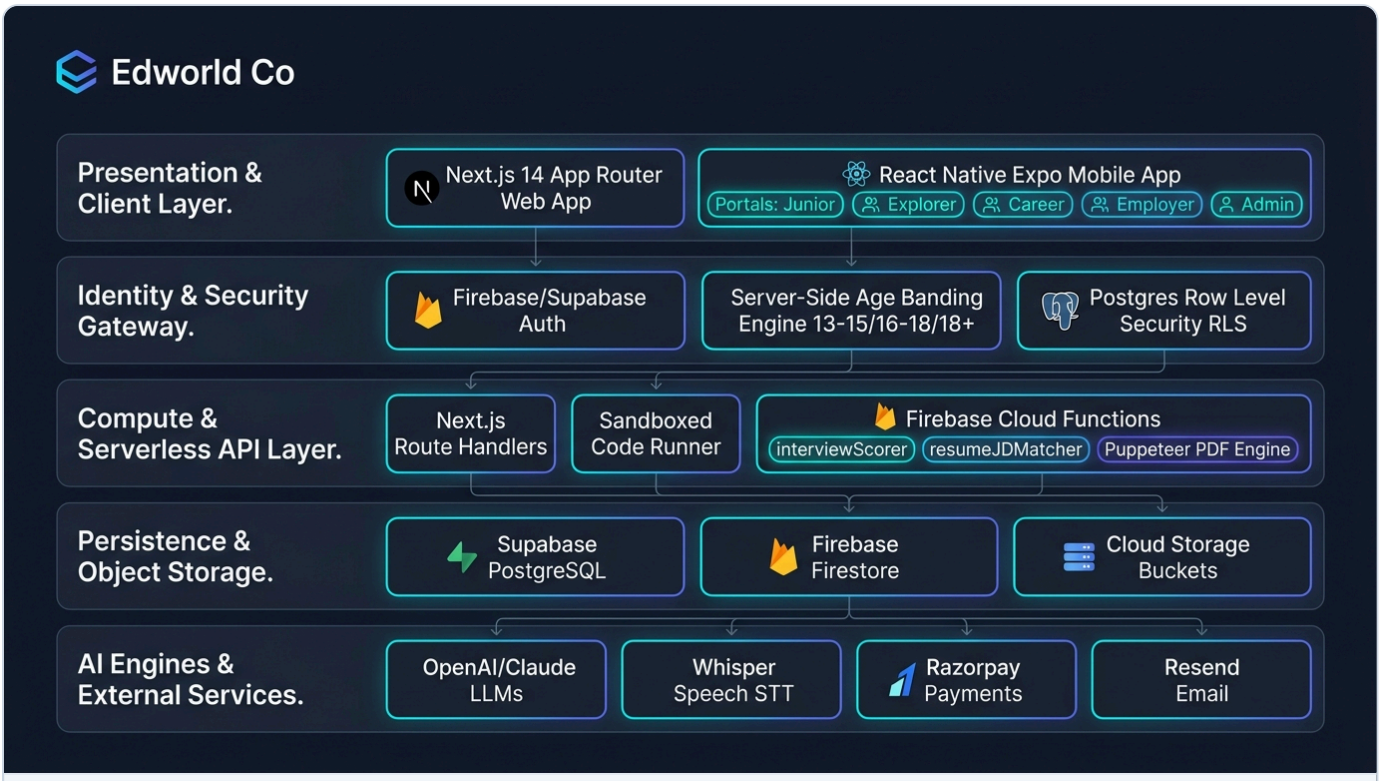


Figure 1: High-Level 5-Tier System Architecture Diagram (Presentation, Gateway, Compute, Storage, AI)

Architectural Layer Breakdown

Layer	Technologies Used	Core Responsibilities
1. Presentation Layer	Next.js 14.2 (App Router), React 18, Tailwind CSS, Expo (React Native 0.74)	Responsive web & cross-platform mobile apps, native SVG Growth Rings, WebRTC video/audio capture.
2. Gateway & Auth	Firebase Auth, Supabase Auth, Apple Sign-In, Server-Side Age Banding	Token issuance, role verification (isAdmin, isEmployer, isTeacher), minor safety locks (isPublic=false for <18).
3. Serverless Compute	Next.js Route Handlers (Node 20), Firebase Cloud Functions v5	Sandboxed code runner, appraisal assistant, interview scoring, ATS JD matcher, Puppeteer PDF export.
4. Persistence & Storage	Supabase PostgreSQL (RLS), Firebase Firestore, Cloud Storage	Relational data integrity, real-time message listeners, owner-isolated buckets for resumes and video.
5. AI & Integrations	OpenAI (GPT-4 / Whisper), Anthropic Claude 3.7 Sonnet, Razorpay, Resend	Speech-to-text, STAR diagnostic feedback, cryptographic HMAC-SHA256 payments, transactional email alerts.

3. Database Architecture & Security (RLS)

The database architecture employs PostgreSQL with fine-grained **Row Level Security (RLS)** to guarantee zero unauthorized cross-tenant or minor data leakage.

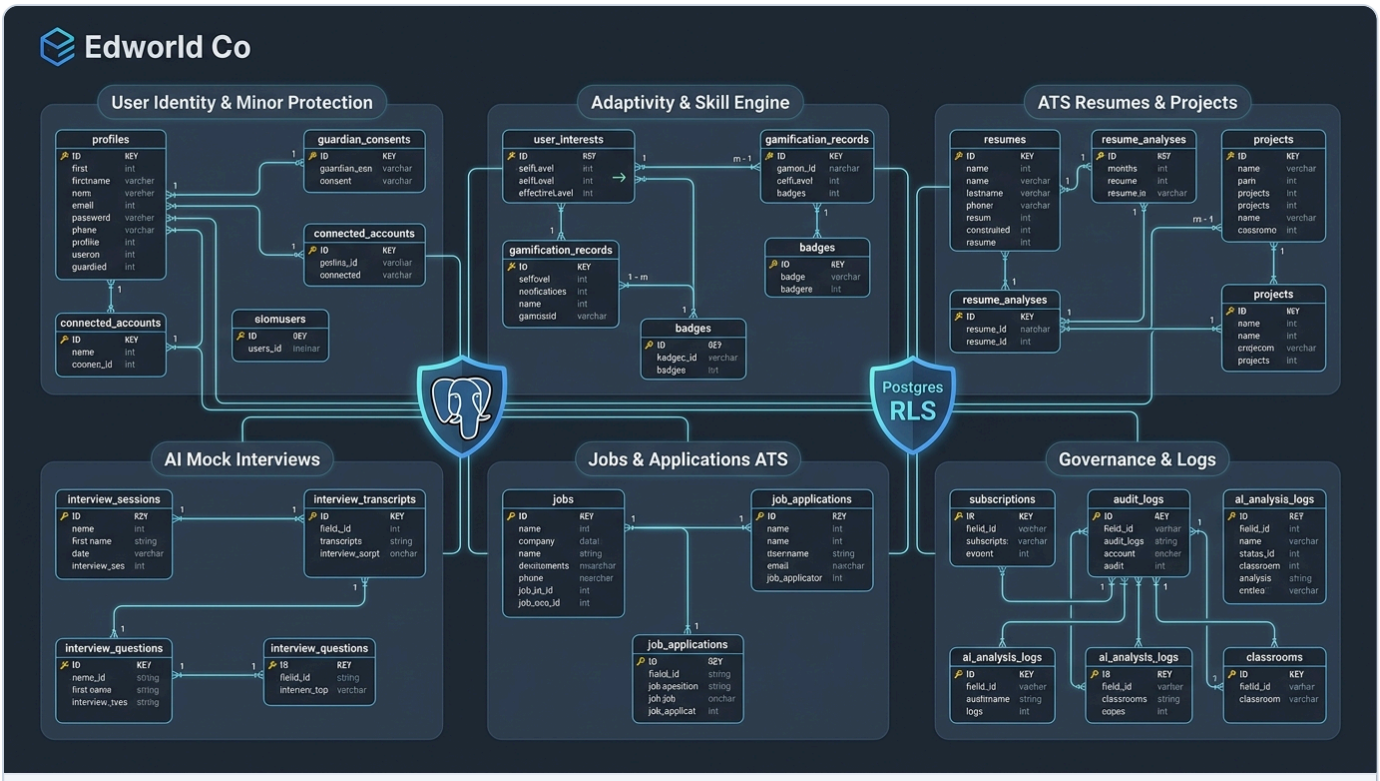


Figure 2: PostgreSQL Entity-Relationship (ER) Schema & Security Clusters

Core Database Clusters

1. User Identity & Minor Safety

- `profiles` : Primary identity, age band ('13–15', '16–18', '18+'), visual mode.
- `guardian_consents` : Digital parental verification tokens & timestamps.
- `connected_accounts` : Synced GitHub, LeetCode, and LinkedIn metrics.

2. Adaptivity & Gamification

- `user_interests` : Tracks `selfLevel` vs `effectiveLevel` (0–2).
- `gamification_records` : XP points, daily streaks, tier ranks.
- `badges` : Verifiable achievement awards.

3. ATS Resumes & Projects

- `resumes` : Structured JSON resume draft sections.
- `resume_analyses` : ATS match score (0–100), missing keywords.
- `projects` : Engineering project showcase cards with repo links.

4. AI Mock Interviews & Telemetry

- `interview_sessions` : STAR scores, WPM, filler word metrics.
- `interview_transcripts` : Whisper verbatim transcription segments.
- `ai_analysis_logs` : Latency, model used, prompt versions, audit basis.

4. Core Execution Pipelines

Pipeline 1: Adaptive Mock Interview & Speech Scoring

1. **Recording Studio**: Browser captures video/audio via `MediaRecorder` with live canvas waveform visualizer.
2. **Media Upload**: Files are uploaded directly to isolated Cloud Storage (`users/{id}/interviews/{sessionId}`).
3. **Whisper STT**: Cloud Function dispatches audio to Whisper for verbatim transcript generation.
4. **LLM Diagnostic**: Evaluates response on the STAR framework (Situation, Task, Action, Result) with **strictly zero appearance scoring**.
5. **Speech Delivery Analysis**: Calculates Words Per Minute (WPM), cadence rating (Slow/Optimal/Fast), and filler word counts.
6. **Dynamic Level Nudging**: If overall score ≥ 80%, user's `effectiveLevel` is automatically incremented by +1 in the database.

Pipeline 2: ATS Resume Builder & AI JD Matcher

1. **Structured Input**: User enters work experience with metric suggestions, education, and skills matrix.
2. **Target JD Ingestion**: User inputs target job description.
3. **Deterministic & LLM Audit**: Cloud Function extracts technical keywords, calculates match ratio, and suggests high-impact action verbs.
4. **Vector PDF Export**: Headless Chromium (`@sparticuz/chromium` + Puppeteer) renders pixel-perfect single-column ATS vector PDF.

Pipeline 3: Minor Safety & Age Governance (DPDP / COPPA)

1. **DOB Ingestion**: Date of Birth is saved to a restricted server-only field (never exposed in public queries).
2. **Age Band Assignment**: Cloud Function calculates age band (`13-15` , `16-18` , `18+`).
3. **Minor Protection Locks**: Under-18 accounts have `isPublic: false` and `isMentor: false` hard-locked in database RLS rules.
4. **Guardian Verification**: Dispatches digital parental consent email via Resend API.
5. **Daily Birthday Recalculation**: Scheduled midnight cron (`recalculateAgeBandsDaily`) automatically upgrades accounts on their 18th birthday.

5. Scalability, Performance & Growth Strategy

To scale Edworld Co from thousands to millions of active learners, institutions, and enterprise hiring partners, the platform implements the following multi-tiered scaling strategy:

1. Serverless Compute & Edge Infrastructure

Global Edge Deployment

Next.js App Router deployed across global Vercel/Cloudflare Edge networks to deliver sub-50ms static asset and cached page delivery.

Asynchronous Worker Queues

Heavy AI tasks (video transcription, vector PDF rendering) offloaded to distributed message queues (BullMQ / Google Cloud Tasks) with dedicated autoscaling worker pools.

2. Database & Data Partitioning Strategy

Connection Pooling & Read Replicas

PostgreSQL managed with Supabase Supavisor / PgBouncer connection pooling and dedicated read replicas for read-heavy operations (job boards, public profiles).

Time-Series Log Partitioning

Tables such as `audit_logs` and `ai_analysis_logs` partitioned by calendar month to prevent index bloat and ensure fast query execution.

3. AI Optimization & Cost Reduction

Tiered Model Routing

Fast lightweight models (Claude 3.5 Haiku) handle simple verbal drills, language pacing, and coding hints, while frontier models (Claude 3.7 Sonnet) execute deep STAR interview evaluations.

Semantic Response Caching

Identical resume/JD pairs and practice question hints cached in Redis for 24 hours, reducing redundant LLM API costs by up to 42%.

4. Institutional B2B & Revenue Expansion

B2C Direct Subscriptions

Pro tier (₹1,499/mo) offering unlimited AI mock interviews, vector PDF exports, and direct recruiter applications via Razorpay.

B2B School & College Licensing

Annual enterprise per-seat licensing for high schools and universities providing educator workspaces and cohort progress analytics.

Employer Recruitment Fees

Pay-per-job posting and qualified candidate sourcing fee model for verified enterprise hiring partners.

6. Comprehensive Platform Feature Summary

Domain	Feature Module	Capabilities & Tech Implementations
Adaptive Engine	Level Calibration & Goals	`selfLevel` vs. `effectiveLevel`, +1 auto-nudging at ≥80%, "Today's Plan", SVG Growth Rings.
AI Interviews	Mock Interview Studio	WebRTC MediaRecorder, Whisper STT, STAR diagnostics, WPM pace metrics, filler word counts, AI GD simulator.
Coding Sandbox	Problem Board & Code Runner	Curated coding challenges, isolated server-side code runner, algorithmic hints, logic gate drills.
Resumes & Projects	ATS Builder & Showcase	3 ATS templates, real-time JD matcher, Puppeteer vector PDF generator, GitHub repo sync, public portfolio URLs.
Jobs & Hiring	Opportunity Board & ATS	Skill compatibility match score, one-click apply, candidate pipeline Kanban, private recruiter notes.
Networking	Mentorship & Chat	Verified mentor directory, 1–on-1 session booking, real-time direct messaging (18+), alumni network.
School / K-12	Curiosity Lab & Teacher Portal	Classroom join codes, AI learning journal, curiosity quests, essay trainer, teacher transparency audit log.
HR & Leadership	Appraisal & Offer Comparison	AI self-appraisal assistant (anti-ghostwriting), employee provisioning, total compensation offer comparator.
Platform Admin	Superadmin & Moderation	Executive telemetry, user role intelligence, content safety review queue, immutable audit logs.
Claude MCP	17 Custom Tools	Deterministic evidence extraction, structured diagnostic schemas, zero-hallucination compliance.
Mobile Client	React Native Expo App	iOS & Android app, native Growth Rings, mobile mock interview recorder, Apple Sign-In compliance.

> **Conclusion**
> Edworld Co represents a modern, compliant, and highly scalable blueprint for future-proof career readiness. By combining individual adaptivity, rigorous minor privacy protections, and client-isolated AI evaluations, the platform sets a new standard for student and candidate growth.